

Question 1

2 marks 101

A group of 7 friends decide to go to a movie.

Determine how many ways the friends can sit in a row if two of the friends refuse to sit next to each other.

Question 2

2 marks 102

Gabrielle listens to her radio at a sound level of 80 dB. She attended a music concert that had a sound level of 115 dB. Determine how many times more intense the music concert was than the radio.

You may use the formula:

$$\beta = 10 \log \left(\frac{I}{I_0} \right)$$

where β is the intensity level of sound, measured in dB

I is the intensity of sound

I_0 is the standard minimum intensity that a person can hear

Question 3

4 marks 103

Solve, algebraically.

$$2(7)^x = 3^{2x-3}$$

Question 4

4 marks 104

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta + 2 \csc \theta - 8 = 0$$

Question 5

1 mark 105

You have forgotten the code to unlock your cell phone. You know the code is made up of four numbers from 0 to 9.

Determine the number of possible codes, if repetition is allowed.

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks 106

In the binomial expansion of $\left(\frac{7}{x^3} - 3x^7\right)^n$, the 5th term contains x^7 .

Determine the value of n .

Question 7

1 mark

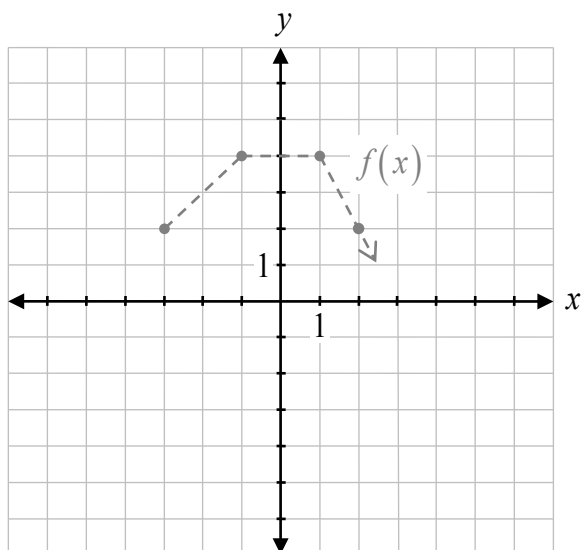
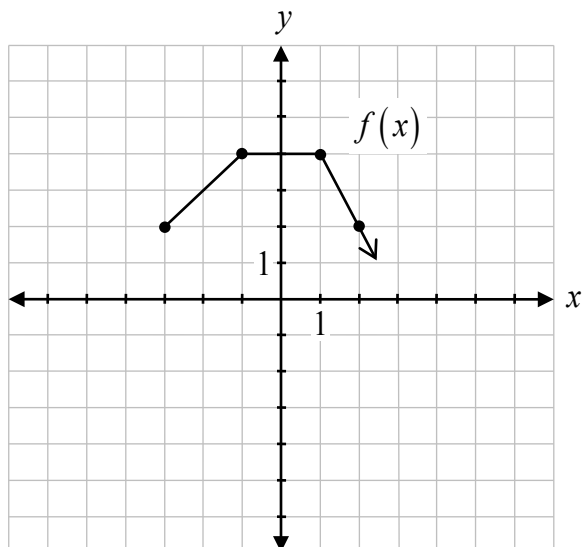
107

Given the domain of $f(x)$ is $\{-6, 1, 3, 4\}$ and the range of $f(x)$ is $\{-4, 7, 10, 15\}$, state the domain of $f^{-1}(x)$.

Question 8

1 mark 108

Given the graph of $y = f(x)$, sketch the graph of its inverse.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 9

3 marks

109

Prove the following identity for all permissible values of θ .

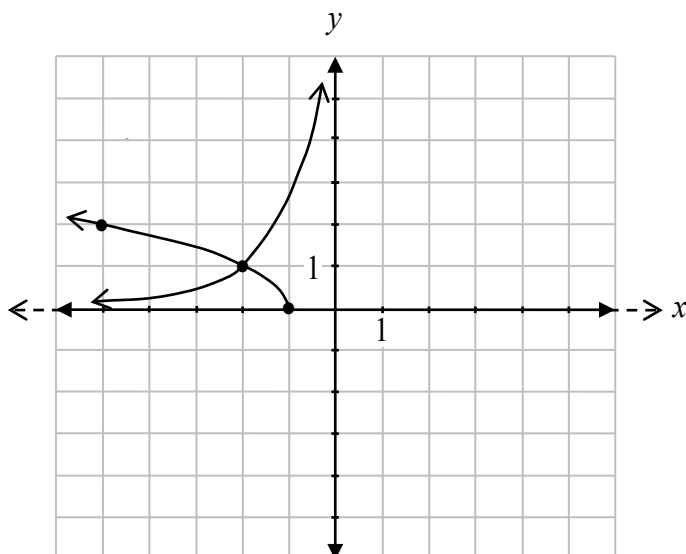
$$\frac{1 + \cos \theta}{1 - \sin^2 \theta} = \sec \theta + \tan^2 \theta + 1$$

Left-Hand Side	Right-Hand Side

Question 10

1 mark 110

Thomas used graphs to solve the equation $e^{x+2} = \sqrt{-(x+1)}$.



He incorrectly states the solution as $(-2, 1)$.

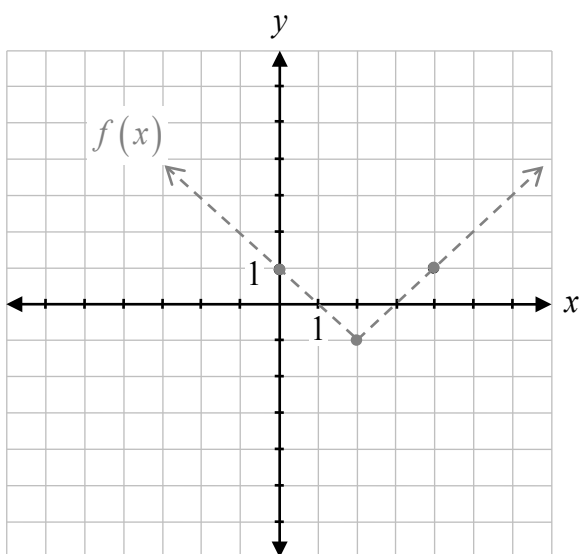
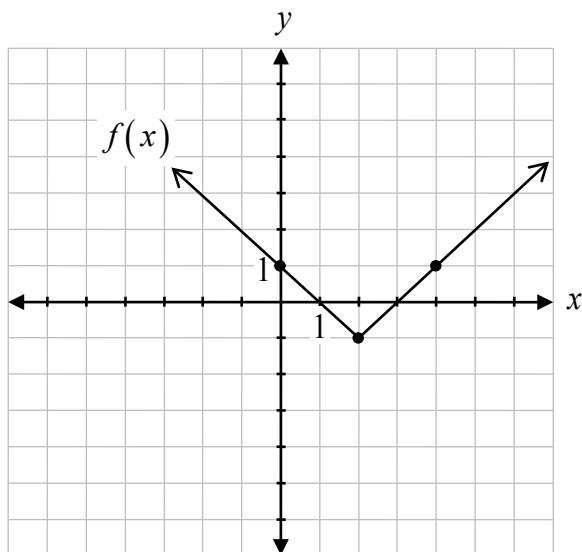
Describe how Thomas should have stated the solution.

Question 11

2 marks

111

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 12

a) 1 mark b) 1 mark

112
113

When a polynomial, $P(x)$, is divided by $(x - 2)$ the resulting equation is

$$\frac{P(x)}{x - 2} = x^2 - x + 1 + \frac{3}{x - 2}.$$

a) Explain why $x - 2$ is not a factor of $P(x)$.

b) Determine the equation for the polynomial function $P(x)$.

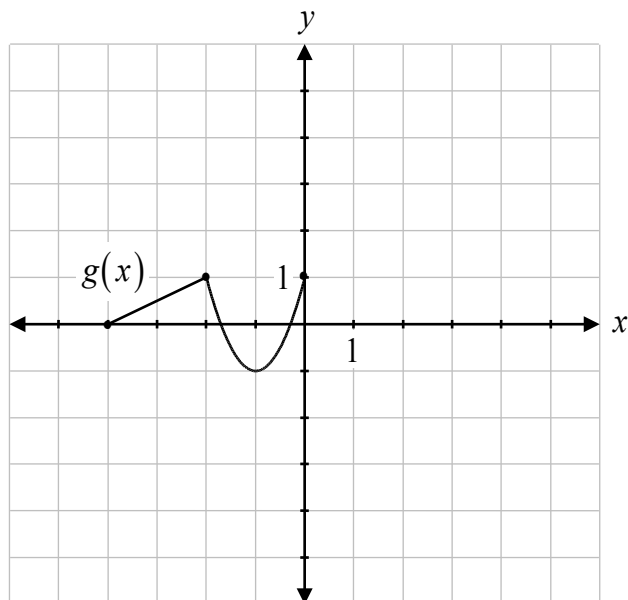
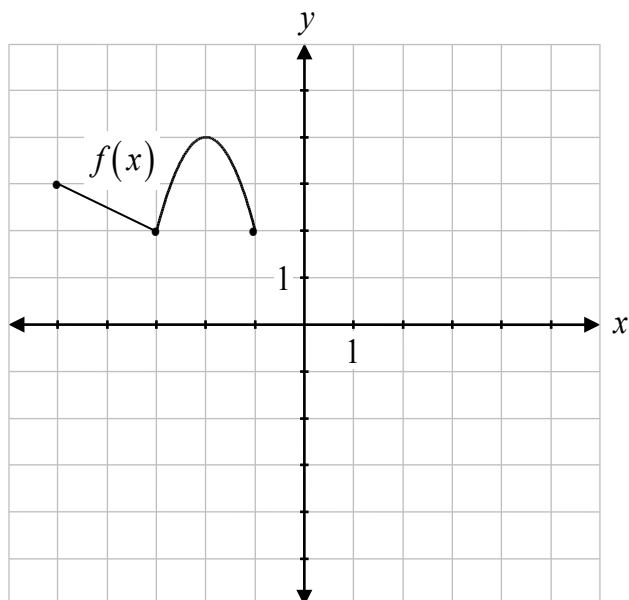
$$P(x) = \underline{\hspace{10cm}}$$

Question 13

3 marks

114

Determine the equation for $g(x)$ in terms of $f(x)$.



$g(x) =$ _____

Question 14

1 mark

115

Explain why the binomial expansion of $(2x + y)^9$ does not have a middle term.

Using the laws of logarithms, completely expand the expression $\log\left(\frac{5\sqrt{a}}{b^3}\right)$.

No marks will be awarded for work done on this page.

